UNIT 2 · 8 hours

# Standard Test Signals



Time Domain Analysis of Control Systems

**Control Systems (22EEG53)**
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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# Learning objectives

- Define step, ramp, parabolic, impulse and sinusoidal test signals
- Write their Laplace transforms
- Explain why standard signals are used
- Relate the signals to each other and to the transient and steady-state response

**Syllabus context**

Time domain analysis of control systems: standard test signals, time response of zero, first and second order systems, time domain specifications, type of systems, static error constants and steady-state error, generalized error series.

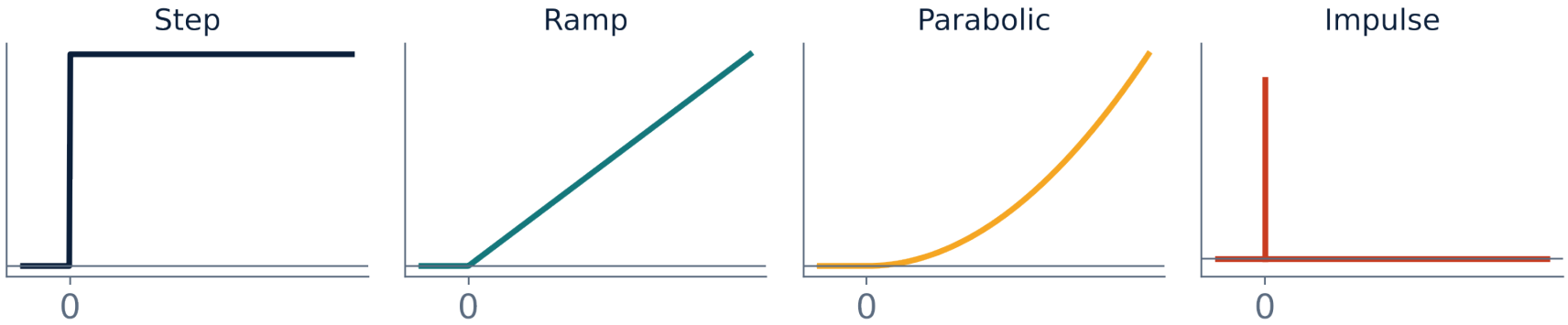
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# Why standard test signals?

- Actual inputs are often unknown or random.
- A few simple signals let us compare systems on an equal basis.
- They represent common real inputs: sudden change (step), constant velocity (ramp), constant acceleration (parabola), shock (impulse).
- System response to these signals indicates behaviour for other inputs (for LTI systems).

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# The four common signals




*Step, ramp, parabolic and impulse inputs*

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# Definitions and Laplace transforms

| Signal | r(t) for t ≥ 0 | R(s) |
|---|---|---|
| **Unit impulse δ(t)** | ∞ at t = 0, area 1 | 1 |
| **Unit step u(t)** | 1 | 1/s |
| **Unit ramp** | t | $1/s^2$ |
| **Unit parabola** | $t^2/2$ | $1/s^3$ |
| **Sinusoid** | A sin ωt | $A\omega/(s^2 + \omega^2)$ |

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# Physical meaning

### Step

Sudden change: switching on a supply, a change in set-point, setting a thermostat. Tests transient behaviour and steady-state accuracy.

### Ramp

Constant velocity input: tracking a moving target at constant speed. Tests velocity error.

### Parabola / impulse

Constant acceleration input; and a sudden shock or disturbance. The impulse response is the transfer function in the time domain: $g(t) = L^{-1}\{G(s)\}$.

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# Transient and steady-state response

**Total response:** $c(t) = c_{tr}(t) + c_{ss}(t)$.

**Transient:** part that dies out as $t \to \infty$; shows speed and oscillation of response.

**Steady state:** part remaining as $t \to \infty$; shows accuracy.

**Absolute stability:** required: the transient must decay.

**Specifications** are set for both: e.g. overshoot and settling time (transient), error (steady state).

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# Key takeaways

- Standard signals: impulse, step, ramp, parabola (and sinusoid)
- Laplace transforms: 1, 1/s, $1/s^2$, $1/s^3$
- Each signal is the integral of the previous one
- Response = transient + steady state

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# Practice questions

**Q1.** Write the Laplace transforms of unit step, ramp and parabola.

**Q2.** Why are standard test signals used?

**Q3.** Define transient and steady-state response.

**Q4.** Which test signal is used to find the velocity error?